

FULL-STACK API ENGINEERING — COURSE OUTLINE (10 MODULES)
DIGITAL COURSE
Piezo-Pico to Femtotechnology Sensors Inc.
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Module 1 — API Foundations & Modern Architecture

- What APIs are (REST, GraphQL, FAST, RPC, Webhooks)
- HTTP lifecycle (request → routing → controller → response)
- JSON serialization/deserialization
- API versioning strategies
- API documentation (OpenAPI/Swagger)
- Modern API patterns (microservices, serverless, edge functions)
- Schematic: Device → Router → ISP → Cloud → API → ISP → Router → Device

Module 2 — Backend Architecture & Folder Structure (FastAPI)

- Production backend folder structure
- Routers, models, schemas, services
- Environment variables & configuration
- Logging setup
- Database connection & dependency injection
- Backend request lifecycle
- How backend structure transforms frontend behavior

Module 3 — Frontend Integration: How Backend Transforms Frontend

- Frontend folder structure (React/Next.js)
- API service layer (Axios/Fetch)
- Transforming backend JSON into UI
- Protected routes
- Error & loading states
- Real-time updates (WebSockets)
- Frontend request lifecycle

Module 4 — Authentication & Identity Systems

- JWT authentication (backend)
- Firebase Authentication
- GitHub OAuth
- Stripe authentication for paid content
- Role-based access (admin/student)
- Protecting backend & frontend routes
- Debugging auth failures

Module 5 — Testing, Observability & Quality Assurance

- Unit tests (services, logic)
- Integration tests (API endpoints)
- Frontend component tests
- Logging & observability
- Performance testing (latency, throughput)
- Debugging full-stack issues

Module 6 — Databases & Persistence Layer

- SQLAlchemy ORM
- Database folder structure
- CRUD operations
- Relationships (1-to-many, many-to-many)
- Alembic migrations
- Query optimization
- Database testing

Module 7 — Payments, Webhooks & External APIs

- Stripe checkout integration
- Stripe webhook processing
- GitHub OAuth login

- Firebase token verification
- Calling external APIs (GitHub, SendGrid, Twilio)
- Securing external API keys
- Debugging external API failures

Module 8 — Deployment & Cloud Infrastructure

- Dockerizing the backend
- Deploying backend to Render/Railway/Azure/AWS
- Deploying frontend to Vercel
- Environment variables in production
- CI/CD pipelines (GitHub Actions)
- Monitoring & logging in production
- Debugging deployment failures

Module 9 — Modern API Enhancements

- Redis caching
- Background tasks
- File uploads (images, PDFs, slides)
- Rate limiting
- Analytics logging
- Async performance
- Response optimization

Module 10 — Final Project (Full-Stack API System)

- Build full backend folder structure
- Build full frontend folder structure
- Implement authentication (JWT + Firebase/GitHub)
- Implement Stripe payments + webhook
- Implement database + migrations
- Implement modern enhancements

- Debug full-stack issues
- Deploy backend + frontend
- Present final system architecture

Publication & Contact Information

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